Midterm Exam 2 Formula Sheet (Exam covers Ch6-9)

Chapter 3

Displacement: $\Delta x = x_f - x_0$

Average velocity: $\bar{v} = \frac{x_f - x_0}{t_f - t_0}$

Instantaneous velocity: $v = \frac{dx}{dt}$

Average acceleration: $\bar{a} = \frac{v_f - v_0}{t_f - t_0}$

Instantaneous acceleration: $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

Kinematic equations:

$\bar{v} = \frac{v + v_0}{2}$

$v = v_0 + at$

$x = x_0 + v_0 t + \frac{1}{2}at^2$

$v^2 = v_0^2 + 2a(x - x_0)$

Acceleration due to gravity:

$a = -g = -9.80 m/s^2$

Chapter 4

Kinematic equations for projectile motion:

$x = x_0 + v_{0x} t$

$v_y = v_{0y} - gt$

$y = y_0 + v_{0y} t - \frac{1}{2}gt^2$

$v_y^2 = v_{0y}^2 - 2g(y - y_0)$

Maximum height: $h = \frac{v_0^2 \sin^2\theta_0}{2g}$

Horizontal range: $R = \frac{v_0^2 \sin 2\theta_0}{g}$

Total time of flight: $t = \frac{2v_0 \sin\theta_0}{g}$

Uniform circular motion:

$a_c = \frac{v^2}{r}$

$T = \frac{2\pi r}{v}$

Chapter 5

Newton's second law of motion: $\vec{F}_{net} = m\vec{a}$

Component form of Newton's second law:

$\Sigma F_x = ma_x$

$\Sigma F_y = ma_y$

$\Sigma F_z = ma_z$

Static friction: $f_s \leq \mu_s N$

Kinetic friction: $f_k = \mu_k N$

Spring force: $F = -k\Delta x$

Chapter 6

Centripetal force: $F_{net} = ma_c = m\frac{v^2}{r}$

Universal gravitation: $F_g = G\frac{m_1 m_2}{r^2}$,

where $G = 6.67 \times 10^{-11}\ N \cdot m^2/kg^2$

Chapters 7 & 8

Work done by a constant force: $W = \vec{F} \cdot \Delta\vec{r} = F\Delta r \cos\theta$

Work done by a varying force: $W = \int_{x_i}^{x_f} F_x dx$

Kinetic Energy: $K = \frac{1}{2}mv^2$

Gravitational potential energy: $U_g = mgh$

Elastic potential energy: $U_s = \frac{1}{2}kx^2$

Work-energy theorem: $W_{ext} = \Delta K = K_f - K_i$

Conservation of energy:

$$\Delta E = \Delta K + \Delta U = W_{ext/nc}$$

Average power: $P_{avg} = \frac{W}{\Delta t}$

Instantaneous power: $P = \vec{F} \cdot \vec{v}$

Chapter 9

Linear momentum: $\vec{p} = m\vec{v}$

Impulse: $\vec{J} = \int_{t_i}^{t_f} \vec{F}_{net} dt = \vec{F}_{net,avg}\Delta t$

Impulse-momentum theorem: $\Delta\vec{p} = \vec{J} = \vec{F}_{avg}\Delta t$

Conservation of momentum: $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$

Elastic collision:

$$\frac{1}{2}m_1 v_{1i}^2 + \frac{1}{2}m_2 v_{2i}^2 = \frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2 v_{2f}^2$$

Center of mass: $x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$

Extra equations

Trigonometry:

$\sin\theta = \frac{opposite}{hypotenuse}$

PHY 2048 Studio

$\cos\theta = \frac{adjacent}{hypotenuse}$

$\tan\theta = \frac{opposite}{adjacent}$

Solutions to the quadratic equation

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Integral: $\int x^n dx = \frac{x^{n+1}}{n+1}$